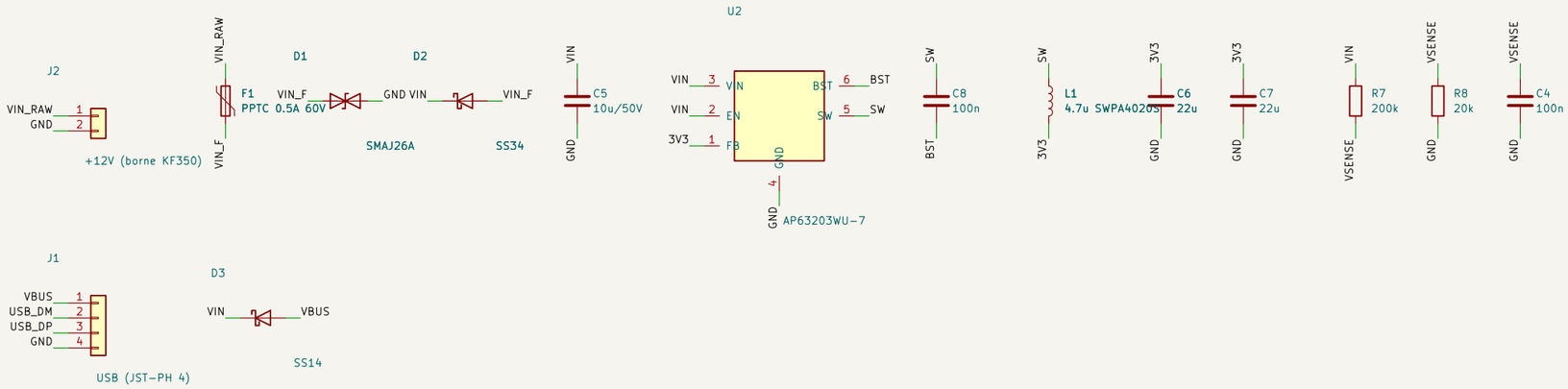
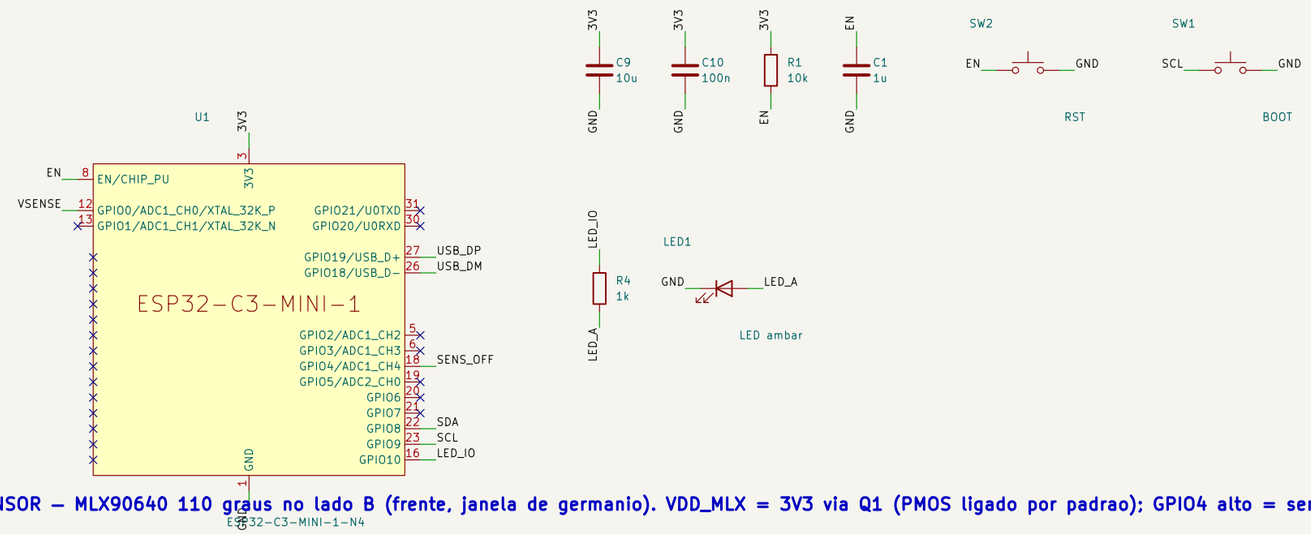
SAFEBOAT TERMICA v1 — camera termica MLX90640 + ESP32-C3, 12 V do barco · esquematico p/ revisao

Redes por label (mesma netlist pino a pino de make_board.py). Placa quadrada 32 x 32 mm; sensor no lado B (frente), resto no lado F (re). Montagem completa JLCPCB (2 lados).

1 · ALIMENTACAO — +12 V (borne J2) -> F1 PPTC 0,5 A -> D1 TVS SMBJ26A -> D2 SS34 (inversao) -> VIN -> U2 AP63203 (EN=VIN, FB=VOUT) -> L1 4,7 uH -> 3V3 · USB J1 -> D3 SS14 -> VIN (bancada) · R7/R8 leem VIN no ADC (GPIO0)



2 · MCU — ESP32-C3-MINI-1: GPIO8 SDA · GPIO9 SCL (+BOOT) · GPIO10 LED · GPIO4 SENS_OFF · GPIO0 VSENSE · GPIO18/19 USB · EN com R1/C1 + RST



3 · SENSOR — MLX90640 110 graus no lado B (frente, janela de germanio). VDD_MLX = 3V3 via Q1 (PMOS ligado por padrao); GPIO4 alto = sensor desligado. Pull-ups I2C p/ VDD_MLX.

